Blastocyst preimplantation genetic diagnosis (PGD) of a mitochondrial DNA disorder

Nathan R. Treff, Ph.D.,[a,b,c] Jessyca Campos, M.S.,[a,b] Xin Tao, M.S.,[a] Brynn Levy, Ph.D.,[d] Kathleen M. Ferry, B.S.,[a] and Richard T. Scott Jr., M.D.[a,b]

[a] Reproductive Medicine Associates of New Jersey, Morristown; [b] Division of Reproductive Endocrinology and Infertility, Department of Obstetrics, Gynecology, and Reproductive Science, University of Medicine and Dentistry of New Jersey, Robert Wood Johnson Medical School, New Brunswick; [c] Department of Genetics, Rutgers University, Piscataway, New Jersey; and [d] Department of Pathology, College of Physicians and Surgeons, Columbia University, New York, New York

Objective: To evaluate the utility of trophectoderm biopsy for preimplantation genetic diagnosis (PGD) of mitochondrial (mt) DNA mutation load.
Design: A PGD case and analysis of blastocyst mosaicism.
Setting: Academic center for reproductive medicine.
Patient(s): A 30-year-old carrier of 35% 3243A>G mtDNA mutation load with a daughter affected by mitochondrial encephalomyopathy, lactic acidosis, and stroke-like episodes (MELAS) syndrome.
Intervention(s): Blastocyst biopsy for PGD of mutation load and gender.
Main Outcome Measure(s): Variation in mutation load within and among embryos, and newborn mutation load after PGD-based selection.
Result(s): Oocytes and embryos were found to possess a variety of 3243A>G mutation loads from 9% to 90% in oocytes and 7% to 91% in embryos, demonstrating that PGD would be a relevant procedure. Highly consistent results were obtained within multiple biopsies of both cleavage- and blastocyst-stage embryos. Importantly, mutation loads observed in trophectoderm were predictive of the inner cell mass ($r^2 = 0.97$). Transfer of a male embryo, predicted to possess 12% mutation load by analysis of a trophectoderm biopsy, resulted in the delivery of a boy with tissue-specific mutation loads ranging from undetectable to 15%.
Conclusion(s): This study represents the first successful clinical application of PGD to reduce the transgenerational risk of transmitting an mtDNA disorder and supports the applicability of blastocyst trophectoderm PGD for carriers of mtDNA mutations attempting reproduction. (Fertil Steril® 2012;98:1236–40. ©2012 by American Society for Reproductive Medicine.)
Key Words: Mitochondrial DNA, MELAS, preimplantation genetic diagnosis, mutation load

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Carriers of mitochondrial DNA disorders have limited reproductive options to ensure the heath of their biological offspring. Mitochondrial DNA mutation loads can vary widely among carriers, and within tissues of individual carriers, making it challenging to manage clinically. Nuclear transfer, cytoplasmic transfer, and preimplantation genetic diagnosis (PGD) have been proposed as methods to prevent transmission (1–3). Although PGD is the only method that has been applied clinically, experience remains limited (4–6). This is largely due to the uncertainty of whether PGD always represents an effective solution (7). For example, mutation load thresholds for which clinical manifestation of the disease can be expected have often not been clearly defined. This is particularly true for mitochondrial encephalomyopathy, lactic acidosis, and stroke-like episodes (MELAS) syndrome where individuals with identical mutation loads can manifest differing severity of the disorder (8). Therefore, even PGD-based selection of embryos with relatively low mutation loads cannot guarantee prevention of the disease in the resulting child.

Further complicating the general applicability of PGD to mitochondrial disorders is the lack of knowledge

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Reprint requests: Nathan R. Treff, Ph.D., Reproductive Medicine Associates of New Jersey, 111 Madison Ave, Suite 100, Morristown, NJ 07960 (E-mail: ntreff@rmanj.com).

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regarding the variable inheritance of mutant mitochondria in a specific carrier patient's oocytes and embryos. Previous studies have provided encouraging results which indicate that variable inheritance does occur during oogenesis (3, 9), providing an opportunity for PGD-based selection of low mutation load embryos.

Another important consideration is whether mitochondrial DNA mutation load is consistent throughout the embryo. If variable amounts are present (mosaicism), then biopsy of one portion of the embryo may result in predicting a mutation load that poorly represents the mutation load in the remaining embryo. Some studies have indicated that very little mosaicism exists within cleavage-stage embryos derived from carrier patients (4–6, 9). Because there is growing interest in the evaluation of trophectoderm from the blastocyst for PGD, it may also be important to characterize whether the mutation load found in trophectoderm from the blastocyst is predictive of the inner cell mass. Trophectoderm analysis may avoid the negative impact on implantation observed from blastomere biopsy (10) and the need to perform biopsy more than once in order to obtain a reliable prediction of mutation load (i.e. two blastomeres (5)), because multiple cells are typically present in a trophectoderm biopsy.

It is also important to consider whether the mutation load predicted in embryos selected for transfer by PGD is consistent with the mutation loads observed in the resulting children. Although reported cases of mitochondrial DNA mutation PGD have thus far demonstrated that embryo biopsy mutation loads are consistent with those found in the resulting newborns (4–6), additional supportive experience remains critical to further establish the general utility of PGD for mitochondrial DNA disorders.

Finally, carriers of mitochondrial mutations attempting reproduction may also be concerned with preventing inheritance in the third generation. More specifically, selection of male embryos by PGD could help avoid transgenerational risk because mtDNA is maternally inherited (11).

The present study helps address these issues and represents the first case to incorporate blastocyst-stage screening and gender selection to avoid transgenerational risk of inheritance of a mitochondrial DNA disorder.

MATERIALS AND METHODS

Patient

A 30-year-old, 35% 3243A>G mutation load carrier presented for IVF with PGD in order to prevent inheritance of MELAS in her offspring. More than 80% of patients with MELAS harbor a 3243A>G mutation in the mitochondrial $tRNA^{Leu(UUR)}$ gene (12). Studies have shown that the 3243A>G mutation disrupts the structure and function of the human mitochondrial $tRNA^{Leu(UUR)}$ by destabilizing the native fold and promoting dimerization (13). The MELAS disorder is manifested by alterations of the central nervous system, including seizures, muscle weakness, loss of half the vision field, cortical blindness, neurosensory deafness, sudden headaches, and episodic vomiting (14, 15). The patient was informed of her carrier status after a MELAS syndrome diagnosis was made in her symptomatic 2-year-old daughter who displayed 84% mutation load in her blood. All materials described in the present study were obtained under IRB approval and analysis was performed with patient consent.

3243A > G Mutation Load and Y Chromosome qPCR Assays

Previously described primers and probes (16) were purchased from Applied Biosystems Inc. (Foster City, CA) in order to quantify the 3243A>G mutation load using TaqMan quantitative real-time (q)PCR as recommended by the supplier. Y chromosome TaqMan assays were added to the MELAS qPCR assay primer pool with qPCR analysis performed as previously described (17). The proportion of the mutant mtDNA was calculated from the $\Delta C_T(C_T^{wild-type} - C_T^{mutant})$, using the formula: proportion of mutant $= 1/(1 + \frac{1}{2}\,\Delta C_T)$ as previously described (18).

Analysis of Lymphocytes

Single lymphocytes ($n = 12$) and five-lymphocyte samples ($n = 12$) from the patient's and daughter's peripheral blood were used as a model for a blastomere or trophectoderm biopsy, respectively. As a negative control, single cell samples from a normal fibroblast cell line from Coriell Cell Repository, ID GM00323, were used. Cells were prepared and lysed as previously described (19). To test whether assay variation was biological or technical in nature, the patient's single cells ($n = 12$) were pooled and aliquoted back into volumes of lysate equivalent to the volume in which each individual single cell was originally prepared (single cell equivalent). This process removes biological variation from the lysates because they were pooled from multiple single cells but maintains the single-cell quantity of DNA per sample. A preamplification reaction was performed using TaqMan PreAmp Master Mix as recommended by the supplier (Applied Biosystems Inc.) using 18 cycles. SDs of the mean mutation loads were compared using an F test.

Analysis of Oocytes, Embryos, and Newborn DNA

Unfertilized oocytes were evaluated to determine levels of variation in gametes from the patient. Individual blastomeres from each of the arrested cleavage-stage embryos were randomized, blinded, and evaluated for mutation load to determine variation within each embryo. Trophectoderm biopsies were obtained from each of the developmentally competent blastocyst-stage embryos prior to cryopreservation as previously described (19). These samples underwent analysis of both the mutation load and gender. Blastocysts that were not used clinically were also prepared for further evaluation of intraembryo variation as previously described (20). The predictive value of trophectoderm mutation load for inner cell mass mutation load was evaluated by calculating the Pearson correlation coefficient. Newborn total DNA was isolated from a buccal swab, as previously described (21), for mutation load analysis. Results from an external commercial laboratory performing mutation load

testing at 5 and 12 months of age were also obtained for further confirmation.

RESULTS

Analysis of Lymphocytes

All single- and five-cell samples successfully amplified both wild-type and mutant alleles, indicating 100% reliability of producing a result. Results of mutation load analysis in single- and five-cell samples were consistent with the expected mutation loads from each individual (Fig. 1A), demonstrating maintenance of mutation and wild-type ratios after preamplification. Variation in the patient's single- and five-cell samples were significantly larger than the daughter's ($P=.0138$) (Fig. 1B). The coefficient of variation (CV) of the daughter's single cells was 6%, whereas the patient's single-cell CV was 29.8%. In addition, the daughter's five-cell sample CV was 4.8%, whereas the patient's five-cell sample CV was 52.9%. Single-cell lysates from the patient were pooled and aliquots of single-cell equivalent volumes were made to eliminate biological contribution to variation that may be present in true single cells. Figure 1B shows a significant reduction ($P<.0001$) in the variation observed in single-cell equivalents (2.4% SD and 5.5% CV) compared with the patient's true single cells (9.1% SD and 29.8% CV). This result suggests that the variation observed in the patient's blood was biological and not technical in nature and that tissue-specific variation in mutation load may increase as a function of age because similar variation in the daughter's blood was not observed.

Analysis of Oocytes, Embryos, and Newborn DNA

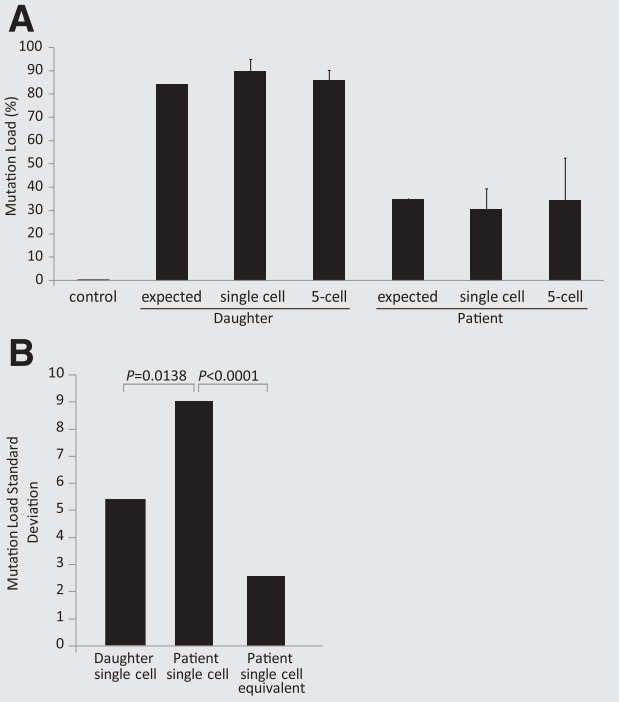
Eight unfertilized oocytes displayed mutation loads ranging from 9% to 90% (Table 1) and eight arrested embryos (six cleavage and two morula) displayed mutation loads ranging from 7% to 86% (Table 1). The SDs of mutation loads of two to seven sister blastomeres from each of five cleavage-stage embryos that arrested ranged from 0.5% to 2.9% (Table 2), indicating faithful representation of the entire embryo from a single blastomere biopsy. Eight developmentally competent blastocysts revealed mutation loads ranging from 12% to 90% (Table 1). Embryo number 7 arrested and was evaluated in whole. Embryo number 8 was selected for transfer as described below. The six remaining blastocysts were reevaluated by thawing and obtaining three to six additional biopsies of both trophectoderm and inner cell mass origin. Overall SDs ranged from 2.1% and 5.0% (Table 2). Trophectoderm biopsy mutation loads were highly concordant with the inner cell mass (Pearson correlation coefficient of 0.97), indicating faithful representation of trophectoderm for the embryo proper (Fig. 2).

The patient elected to transfer the 12% mutation load embryo because it represented both the lowest mutation load of all the available embryos and a gender (male) that would avoid inheritance of the mutation in the third generation. The patient became pregnant and delivered a boy. Newborn buccal cell DNA analyzed with the assay developed in the present study indicated a mutation load of 15% at 1 month of age. In addition, testing results of a commercial laboratory indicated undetectable levels of the mutation from both a buccal swab and urine sediment, and <10% mutation load in the blood at 5 and 12 months of age.

DISCUSSION

An important observation from the present study is that very little mosaicism of mutation load was present at both the cleavage and blastocyst stages of development. This is consistent

FIGURE 1



Single- and five-cell lymphocyte sample mutation loads. (A) Single- and five-cell samples from the daughter and patient with respect to the expected loads of 84% and 35%, respectively. (B) Variation in the single-cell, five-cell, and single-cell-equivalent samples.

Treff. Trophectoderm mtDNA mutation load PGD. Fertil Steril 2012.

TABLE 1

A3243G mutation load at different stages of preimplantation development.

Sample type	Mean ± SD (%)	Individual samples (%)							
		1	2	3	4	5	6	7	8
Oocyte[a] (n = 8)	60 ± 14	60	61	58	68	68	9	89	70
Cleavage (n = 6)	44 ± 31	7[b]	86[b]	71[b]	52[b]	25[b]	24[c]		
Morula (n = 2)	76 ± 21	61[c]	91[c]						
Blastocyst (n = 8)	61 ± 32	24[d]	81[d]	90[d]	89[d]	82[d]	38[d]	74[c]	12[e]
All samples (n = 24)	58 ± 28								

[a] Whole oocytes that failed to fertilize.
[b] Average of multiple blastomeres (see Table 2).
[c] Whole embryo.
[d] Average of multiple biopsies (see Table 2).
[e] Selected for transfer.

Treff. Trophectoderm mtDNA mutation load PGD. Fertil Steril 2012.

TABLE 2

Variation of A3243G mutation load within cleavage- and blastocyst-stage embryos.

Cleavage-stage embryo no.	SD (%)	Blastomere no. (%) 1	2	3	4	5	6	7
1	1.0	7.0	7.6	5.1	6.3	7.9	8.0	7.2
2	0.5	86	85					
3	1.3	68	70	68	70	69	67	67
4	2.9	47	53	53	53	53	56	
5	2.9	26	22	23	26	30	22	

Blastocyst no.	SD (%)	Trophectoderm biopsy no. (%) 1	2	3	4	Inner cell mass biopsy no. (%) 1	2	3
1	3.5	19	27	25		25		
2	5.0	75	86	84		81		
3	3.8	87	90	94	94	85		
4	2.6	88	91	86		91		
5	2.4	79	85	83	80	81	84	85
6	2.1	36	35	38		39	41	38

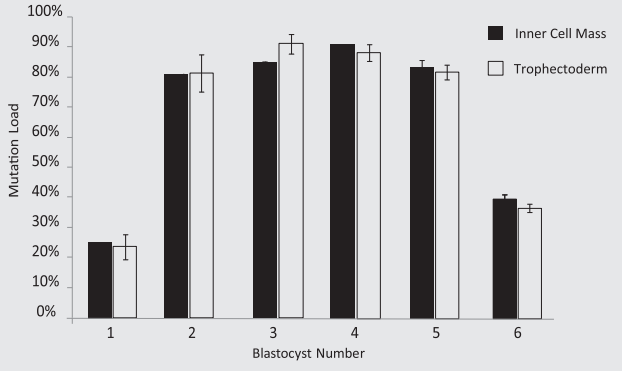
Treff. Trophectoderm mtDNA mutation load PGD. Fertil Steril 2012.

with previous reports that evaluated cleavage-stage embryos (9, 10) but also represents the first study of mosaicism in multiple blastocysts. This study also represents only the fourth reported case of mitochondrial DNA PGD and the second report specific for MELAS. Although 3 different tissues from the newborn were evaluated and displayed <15% mutation load, it is possible that because of the known variation among different tissues within an individual, other untested tissues may possess higher mutation loads. However, as with previously published mtDNA PGD cases where only one or no newborn tissue samples were evaluated (4–6), access to additional tissue is difficult. Although the present study represents an important step toward more generalized use of PGD to prevent transmission of mitochondrial DNA disorders, it is also the first to incorporate gender selection to avoid transgenerational inheritance. The ethics of gender selection in the context of mtDNA mutation load PGD have been more appropriately and extensively reviewed elsewhere. Indeed, Bredenoord et al. (22) has suggested that this approach may only be morally acceptable if it can be done with relatively little additional expense and only as an adjunct to PGD testing already being conducted to determine the mutation load. Indeed, the present study met each of these suggested criteria.

It is expected that the patient's child born from this procedure will not endure the same difficulties with his own efforts to produce healthy children. Still, caution is warranted with respect to generalized conclusions regarding the effectiveness of blastocyst PGD in predicting the mutation load in all cases and in all tissues within resulting newborns. Nonetheless, this study represents an important resource in establishing additional evidence for PGD-based prevention of inheritance of mitochondrial DNA disorders.

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FIGURE 2



Mean (±SD where applicable) mutation loads in trophectoderm and inner cell mass from 6 blastocysts.

Treff. Trophectoderm mtDNA mutation load PGD. Fertil Steril 2012.

REFERENCES

1. Poulton J, Chiaratti MR, Meirelles FV, Kennedy S, Wells D, Holt IJ. Transmission of mitochondrial DNA diseases and ways to prevent them. PLoS Genet 2010;6.
2. Poulton J, Kennedy S, Oakeshoott P, Wells D. Preventing transmission of maternally inherited mitochondrial DNA diseases. Biom J 2009;338:b94.
3. Brown DT, Samuels DC, Michael EM, Turnbull DM, Chinnery PF. Random genetic drift determines the level of mutant mtDNA in human primary oocytes. Am J Hum Genet 2001;68:533–6.
4. Steffann J, Frydman N, Gigarel N, Burlet P, Ray PF, Fanchin R, et al. Analysis of mtDNA variant segregation during early human embryonic development: a tool for successful NARP preimplantation diagnosis. J Med Genet 2006;43:244–7.
5. Monnot S, Gigarel N, Samuels DC, Burlet P, Hesters L, Frydman N, et al. Segregation of mtDNA throughout human embryofetal development: m.3243A>G as a model system. Hum Mutat 2011;32:116–25.
6. Thorburn D, Wilton L, Stock-Myer S. Healthy baby girl born following pre-implantation genetic diagnosis for mitochondrial DNA m.8993t>g mutation. In: Mitochondrial medicine 2009. Capitol Hill: Molecular Genetics and Metabolism; 2009:5–6.
7. Poulton J, Bredenoord AL. 174th ENMC international workshop: Applying Pre-implantation Genetic Diagnosis to mtDNA Diseases: Implications of Scientific Advances 19-21 March 2010, Naarden, The Netherlands. Neuromuscul Disord 2010;20:559–63.
8. Mehrazin M, Shanske S, Kaufmann P, Wei Y, Coku J, Engelstad K, et al. Longitudinal changes of mtDNA A3243G mutation load and level of functioning in MELAS. Am J Med Genet A 2009;149A:584–7.
9. Tajima H, Sueoka K, Moon SY, Nakabayashi A, Sakurai T, Murakoshi Y, et al. The development of novel quantification assay for mitochondrial DNA heteroplasmy aimed at preimplantation genetic diagnosis of Leigh encephalopathy. J Assist Reprod Genet 2007;24:227–32.
10. Treff NR, Ferry KM, Zhao T, Su J, Forman EJ, Scott RT. Cleavage stage embryo biopsy significantly impairs embryonic reproductive potential while blastocyst biopsy does not: a novel paired analysis of cotransferred biopsied and non-biopsied sibling embryos. Fertil Steril 2011;96:S2.
11. Giles RE, Blanc H, Cann HM, Wallace DC. Maternal inheritance of human mitochondrial DNA. Proc Natl Acad Sci U S A 1980;77:6715–9.
12. Goto Y, Nonaka I, Horai S. A mutation in the tRNA(Leu)(UUR) gene associated with the MELAS subgroup of mitochondrial encephalomyopathies. Nature 1990;348:651–3.

13. Wittenhagen LM, Kelley SO. Dimerization of a pathogenic human mitochondrial tRNA. Nat Struct Biol 2002;9:586–90.
14. Pavlakis SG, Phillips PC, DiMauro S, De Vivo DC, Rowland LP. Mitochondrial myopathy, encephalopathy, lactic acidosis, and strokelike episodes: a distinctive clinical syndrome. Ann Neurol 1984;16:481–8.
15. Montagna P, Gallassi R, Medori R, Govoni E, Zeviani M, Di Mauro S, et al. MELAS syndrome: characteristic migrainous and epileptic features and maternal transmission. Neurology 1988;38:751–4.
16. Singh R, Ellard S, Hattersley A, Harries LW. Rapid and sensitive real-time polymerase chain reaction method for detection and quantification of 3243A>G mitochondrial point mutation. J Mol Diagn 2006;8:225–30.
17. Treff NR, Tao X, Su J, Lonczak A, Northrop LE, Ruiz A, et al. Tracking embryo implantation using cell-free fetal DNA enriched from maternal circulation at 9 weeks gestation. Mol Hum Reprod 2011;17:434–8.
18. Bai RK, Wong LJ. Detection and quantification of heteroplasmic mutant mitochondrial DNA by real-time amplification refractory mutation system quantitative PCR analysis: a single-step approach. Clin Chem 2004;50:996–1001.
19. Treff NR, Su J, Tao X, Miller KA, Levy B, Scott RT Jr. A novel single-cell DNA fingerprinting method successfully distinguishes sibling human embryos. Fertil Steril 2009;94:477–84.
20. Northrop LE, Treff NR, Levy B, Scott RT Jr. SNP microarray-based 24 chromosome aneuploidy screening demonstrates that cleavage-stage FISH poorly predicts aneuploidy in embryos that develop to morphologically normal blastocysts. Mol Hum Reprod 2010;16:590–600.
21. Treff NR, Su J, Kasabwala N, Tao X, Miller KA, Scott RT Jr. Robust embryo identification using first polar body single nucleotide polymorphism microarray-based DNA fingerprinting. Fertil Steril 2010;93:2453–5.
22. Bredenoord AL, Dondorp W, Pennings G, De Wert G. Avoiding transgenerational risks of mitochondrial DNA disorders: a morally acceptable reason for sex selection? Hum Reprod 2010;25:1354–60.